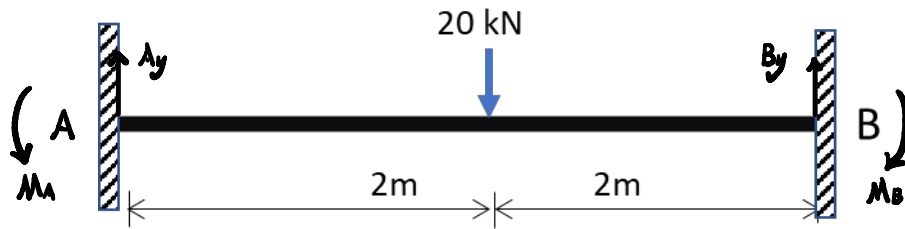


1. Calculate support reactions for the statically indeterminate beams shown below using the double integration method.



$$\sum M_A = M_A - 40 + 4B_y - M_B = 0$$

$$\sum F_y = A_y + B_y - 20 = 0$$

$$M = EI \frac{d^2v}{dx^2} = A_y x - M_A - 20 \langle x-2 \rangle$$

$$\therefore EI \frac{dv}{dx} = \int M dx = \frac{1}{2} A_y x^2 - M_A x - 10 \langle x-2 \rangle^2 + C_1$$

$$\therefore EI v = \int EI \frac{dv}{dx} dx = \frac{1}{6} A_y x^3 - \frac{1}{2} M_A x^2 - \frac{10}{3} \langle x-2 \rangle^3 + C_1 x + C_2$$

$$\text{When } x=0 \text{ and } 4, \quad v=0, \quad \frac{dv}{dx}=0$$

$$\therefore C_1 = 0, \quad C_2 = 0$$

$$\therefore 8A_y - 4M_A - 40 = 0$$

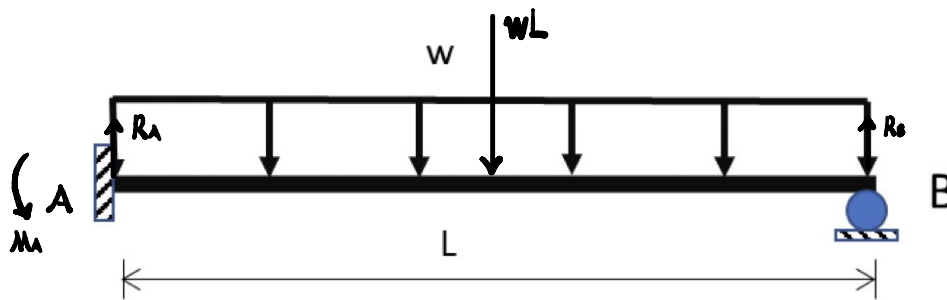
$$\frac{22}{3} A_y - 8M_A - \frac{80}{3} = 0$$

$$\therefore A_y = B_y = 10 \text{ (kN)}$$

$$M_A = M_B = 10 \text{ (kNm)}$$

2. Calculate the supporting reaction at support B, using the unit load method.

a)



$$\sum M_A = M_A - \frac{1}{2}WL^2 + R_B L = 0$$

$$\sum F_y = R_A - WL + R_B = 0 \quad \therefore R_B = WL - R_A \quad \therefore M_A = \frac{1}{2}WL^2 - WL^2 + R_A L = R_A L - \frac{1}{2}WL^2$$

$$EI \frac{d^2v}{dx^2} = M = R_A x - M_A - wL \left\langle x - \frac{L}{2} \right\rangle \quad \frac{wx^2}{2}$$

$$\therefore EI \frac{dv}{dx} = \frac{1}{2}R_A x^2 - M_A x - \frac{1}{2}wL \left\langle x - \frac{L}{2} \right\rangle^2 + C_1 \quad \frac{1}{6}wx^3$$

$$\therefore EI v = \frac{1}{6}R_A x^3 - \frac{1}{2}M_A x^2 - \frac{1}{6}wL \left\langle x - \frac{L}{2} \right\rangle^3 + C_1 x + C_2 \quad \frac{1}{24}wx^4$$

$$\text{When } x=0, v=0, \frac{dv}{dx}=0$$

$$\therefore C_1 = 0$$

$$\therefore C_2 = 0$$

$$\text{When } x=L, v=0$$

$$\therefore \frac{1}{6}R_A L^3 - \frac{1}{2}M_A L^2 - \frac{1}{24}wL^4 = 0$$

$$\therefore \frac{1}{6}R_A L^3 - \frac{1}{2}R_A L^3 + \frac{1}{24}wL^4 - \frac{1}{24}wL^4 = 0$$

$$\therefore -\frac{1}{3}R_A + \frac{1}{24}wL = 0$$

$$\therefore R_A = \frac{5}{12}wL$$

$$\therefore R_B = WL - R_A = \left( w - \frac{5}{12}w \right) L = \frac{7}{12}wL$$

2. Calculate the supporting reaction at support B, using the unit load method.

b)



$$\sum M_A = M_A + LR_B - \frac{3}{2}LP = 0$$

$$\sum F_y = R_A + R_B - P = 0 \quad \therefore R_B = P - R_A \quad \therefore M_A = \frac{1}{2}LP + LR_A$$

$$M = EI \frac{d^2v}{dx^2} = R_A x - M_A + R_B \langle x - L \rangle$$

$$\therefore EI \frac{d^2v}{dx^2} = \frac{1}{2}R_A x^2 - M_A x + \frac{1}{2}R_B \langle x - L \rangle^2 + C_1$$

$$\therefore EI v = \frac{1}{6}R_A x^3 - \frac{1}{2}M_A x^2 + \frac{1}{6}R_B \langle x - L \rangle^3 + C_1 x + C_2$$

When  $x = 0$ ,  $v = 0$  and  $\frac{dv}{dx} = 0$

$$\therefore C_1 = 0$$

$$\therefore C_2 = 0$$

When  $x = L$ ,  $v = 0$

$$\therefore \frac{1}{6}R_A L^3 - \frac{1}{2}M_A L^3 = 0$$

$$\therefore \frac{1}{6}R_A L - \frac{1}{4}PL - \frac{1}{4}R_A L = 0$$

$$\therefore R_A = -\frac{2}{10}P$$

$$\therefore R_B = P - R_A = \frac{12}{10}P$$